Noise-Robust Urban Sound Classification: Evaluating Background Noise and Audio Augmentation Across Deep Learning Models

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***Abstract*—Environmental sound classifiers are commonly evaluated on held-out recordings, yet their behavior under controlled background interference can differ from clean-set performance. This work studies whether training-time audio augmentation improves noise robustness on UrbanSound8K. Three architectures–a compact convolutional neural network (CNN), a convolutional recurrent neural network (CRNN) with a bidirectional gated recurrent unit, and a non-pretrained ResNet18–were each trained in baseline and augmented conditions. All models used the same standardized Log-Mel representation and were evaluated on the same held-out fold under clean, 20 dB, 10 dB, and 0 dB signal-to-noise ratio conditions using deterministic corruptions from a held-out MS-SNSD noise bank. Augmentation increased normalized macro-F1 SNR area under the curve for every architecture. The augmented ResNet18 achieved the strongest overall robustness with a normalized macro-F1 SNR AUC of 0.7258 and a macro F1 of 0.6384 at 0 dB, whereas the baseline CRNN achieved the highest clean macro F1 of 0.8037. The results show that clean-set performance and noise robustness are not identical objectives and that the benefit of augmentation depends on architecture. The study is limited to one official test fold and one completed training seed per configuration, so the findings are reported as controlled empirical evidence rather than a universal robustness claim.**

***Index Terms*—environmental sound classification, audio augmentation, background noise, robustness, Log-Mel spectrogram, UrbanSound8K**

I. Introduction

Environmental sound classification (ESC) aims to identify non-speech acoustic events such as sirens, drilling, engines, horns, and human activity. Public benchmarks such as UrbanSound8K have enabled reproducible comparison of ESC systems and have supported the transition from handcrafted features to learned representations [1], [2]. Subsequent work has shown that deep convolutional models and data augmentation can improve environmental sound recognition, particularly when labeled data are limited [3], [5]. Standard held-out accuracy, however, does not directly describe how rapidly a classifier degrades when the same acoustic event is masked by background interference.

This distinction is practically important because the model that performs best on clean recordings may not retain the most performance under noise. Prior audio-augmentation studies have considered time and pitch modification, mixed examples, and spectrogram masking [3], [4], [7]. The present study therefore focuses on a controlled comparison in which the data split and evaluation protocol remain fixed while training-time augmentation is varied. The analysis examines both whether augmentation reduces noise-related performance loss and whether the size of that effect changes with model architecture.

UrbanSound8K is used as the public data source. A compact 2-D CNN, a CRNN with a bidirectional gated recurrent unit (BiGRU), and a ResNet18 adapted to one-channel Log-Mel spectrograms are each trained in baseline and augmented conditions. The six selected checkpoints are evaluated on identical held-out samples under clean, 20 dB, 10 dB, and 0 dB conditions. Evaluation noise is kept separate from training noise, and corruption is deterministic across models so that architecture-matched comparisons are made on the same inputs.

A. Contributions

The main contributions of this work are:

- a fold-aware and reproducible ESC pipeline with shared preprocessing for three deep architectures;
- an architecture-matched baseline-versus-augmentation experiment using six full training runs and 24 controlled test conditions;
- deterministic SNR-controlled evaluation with disjoint training and test noise banks and identical corruptions across compared models; and
- analysis of clean performance, noisy-condition performance, normalized SNR AUC, augmentation effects, and class-level behavior under severe noise.

II. Literature Review

A. Environmental Sound Classification

UrbanSound8K was introduced with a taxonomy for urban acoustic events and a predefined fold organization intended to support reproducible evaluation [1]. Early deep-learning work showed that convolutional networks trained on time-frequency representations can learn useful spectro-temporal features for ESC and outperform conventional MFCC-based baselines in several settings [2]. Other approaches have learned directly from raw audio, demonstrating that environmental sound recognition can be addressed with either engineered time-frequency representations or end-to-end feature learning [6]. In this study, Log-Mel spectrograms are retained as a common input because they provide a consistent representation across the three compared architectures.

B. Deep Architectures for Acoustic Classification

The three model families used here represent different inductive biases. CNNs operate locally over time-frequency structure and have become a common ESC baseline [2], [3]. Recurrent units provide an additional mechanism for modeling ordered temporal context; the gated recurrent unit was originally introduced as a compact gated recurrent architecture [8]. The CRNN in this work therefore combines convolutional feature extraction with a BiGRU over the resulting temporal sequence. The third model is ResNet18, derived from residual learning architectures that ease optimization of deeper networks through skip connections [9]. Rather than using ImageNet transfer learning, the present ResNet18 is adapted to a single-channel spectrogram input and trained from scratch so that its results do not include external image-domain pretraining.

C. Audio Augmentation and Noise Robustness

Data augmentation has repeatedly improved ESC performance. Salamon and Bello showed that a deeper CNN benefited from an augmented environmental-sound training set and also observed that augmentation effects differed across sound classes [3]. Between-Class learning regularized sound recognition by mixing examples from different classes [4], while later ESC work reported gains from regularization and augmented Log-Mel training data [5]. SpecAugment introduced direct masking of time and frequency regions in filter-bank representations, establishing a simple feature-space augmentation strategy that has since influenced audio model training beyond speech recognition [7].

Table I summarizes the ten sources used to establish the methodological context, including their principal methods, findings, and limitations relative to the present experiment.

The literature supports augmentation as a useful generalization strategy, but benchmark improvement and resistance to additive interference are not the same measurement. Much of the ESC literature reports standard held-out performance, while studies that explicitly introduce noisy conditions often change other parts of the representation or architecture at the same time. This leaves a narrow but useful gap: the contribution of training-time augmentation to noise tolerance is not always isolated under an architecture-matched, leakage-aware protocol. The present work addresses that gap by comparing otherwise matched baseline and augmented versions of three architectures and evaluating them on the same held-out fold with deterministic SNR-controlled corruption. Background noise is drawn from MS-SNSD, which provides environmental noise resources and configurable SNR-based mixing procedures [10].

III. Data Collection, Preprocessing, and EDA

A. Data Source, Inventory, and Split

UrbanSound8K is a publicly available collection of 8,732 labeled excerpts of up to four seconds from ten urban sound classes [1]. The dataset was accessed for this project in August 2026. Its ten predefined folds were preserved instead of constructing a naive random split: folds 1–8 were used for training, fold 9 for validation, and fold 10 for final testing. The test fold was not used for checkpoint selection or hyperparameter decisions.

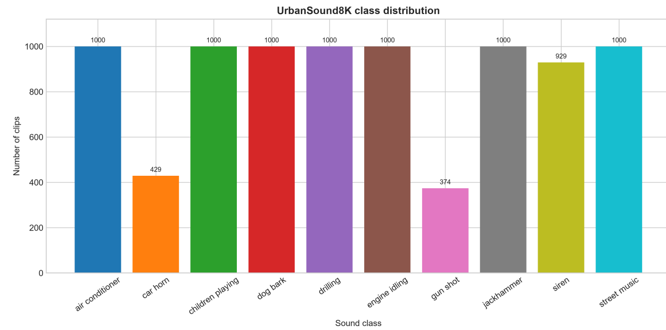

Fig. 1. Class distribution of the complete UrbanSound8K inventory. The smaller car-horn and gun-shot classes motivate macro-averaged evaluation alongside accuracy.

The final split accounts for all 8,732 clips, so no recording was excluded from the experimental inventory. The required filename, label, class identifier, and fold information were available for every retained sample; consequently, no missing-value imputation was required for the metadata used by the pipeline. Because UrbanSound8K contains related excerpts from source recordings, these clips were not randomly redistributed or treated as independent duplicate records. Instead, the official fold organization was retained so that the dataset's source-grouping structure remained intact and leakage risk was reduced.

B. Class Balance and Fold Composition

The class inventory is only approximately balanced. Six classes contain 1,000 clips each, while `car_horn` contains 429 clips, `gun_shot` contains 374, and `siren` contains 929. This imbalance motivated the use of macro-averaged precision, recall, and F1 in addition to accuracy.

The official fold sizes also vary, from 806 clips in fold 8 to 990 in fold 4. The variation is moderate rather than extreme and all folds remain usable as predefined grouped partitions, which supports the decision to preserve the benchmark structure rather than rebalance it through random reassignment.

C. Duration and Recording Characteristics

Clip duration is concentrated near the four-second dataset limit, but shorter events occur frequently in car horn, dog bark, and gun shot. These short events are not removed as outliers because they are legitimate examples of the target classes. Instead, fixed-duration padding is used later so that batch tensors have consistent dimensions without discarding short acoustic events.

Source audio is also heterogeneous in format. The largest groups were recorded at 44.1 kHz (5,370 clips) and 48 kHz (2,502 clips), with smaller numbers at several other rates. Most

TABLE I
SUMMARY OF LITERATURE USED TO POSITION THE STUDY.

Source	**Year**	**Method**	**Key finding**	**Limitation relative to this study**
Salamon *et al.* [1]	2014	UrbanSound taxonomy, benchmark construction, and fold-based evaluation	Established UrbanSound8K as a reproducible benchmark for urban acoustic events	Benchmark-oriented; did not isolate augmentation or controlled noise robustness
Piczak [2]	2015	2-D CNN on time-frequency representations	Demonstrated that learned spectro-temporal features can outperform conventional audio baselines	Small early CNN; emphasis was clean benchmark classification rather than additive-noise robustness
Salamon and Bello [3]	2017	Deep CNN with waveform-level augmentation	Augmentation improved ESC performance and produced class-dependent effects	Did not measure matched-model degradation over a controlled test-time SNR sweep
Tokozume *et al.* [4]	2018	Between-Class learning with mixed audio examples	Mixed-class training regularized deep sound-recognition models	Not designed to isolate robustness to held-out additive background noise
Abdoli *et al.* [6]	2019	End-to-end 1-D CNN on raw waveforms	Showed that useful environmental-sound features can be learned directly from waveform samples	No matched augmentation study and no controlled SNR evaluation
Park *et al.* [7]	2019	SpecAugment time/frequency masking	Simple masking of filter-bank features improved generalization in speech recognition	Developed for ASR rather than UrbanSound8K and does not test additive-noise robustness directly
Reddy *et al.* [10]	2019	MS-SNSD noise resources and configurable SNR mixing	Provided scalable noise data and procedures for controlled noisy-audio generation	Designed around noisy speech, not environmental-sound classification
Mushtaq and Su [5]	2020	Regularized deep CNN with augmented Log-Mel training data	Reported substantial ESC gains from regularization and augmentation	Robustness to progressively stronger held-out noise was not isolated as the main outcome
Cho *et al.* [8]	2014	Gated recurrent unit for sequence modeling	Introduced a compact gated recurrent mechanism for modeling ordered context	General sequence model; not an ESC or noise-robustness study
He *et al.* [9]	2016	Residual networks with skip connections	Residual connections enabled effective optimization of substantially deeper networks	Developed for image recognition; direct applicability to spectrograms requires adaptation

TABLE II
URBANSOUND8K EXPERIMENTAL SPLIT

Subset	Official folds	Samples
Training	1–8	7,079
Validation	9	816
Test	10	837
Total	1–10	8,732

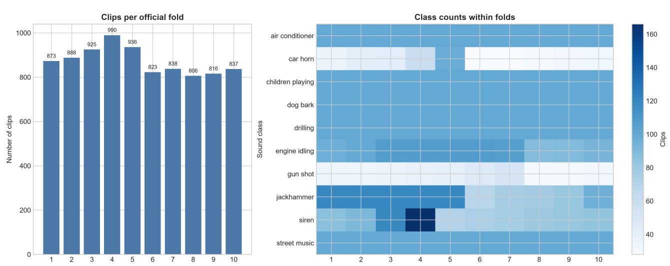

Fig. 2. Number of clips in each official fold and class counts within folds. The predefined partitions are not equal in size, but all ten classes are represented across the evaluation structure.

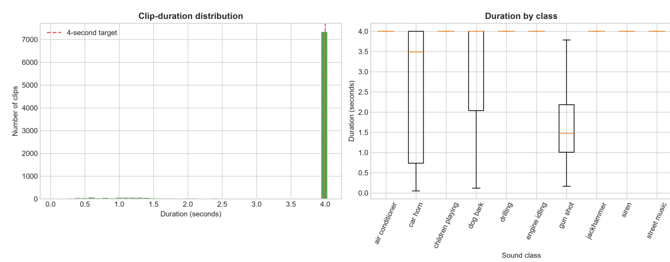

Fig. 3. Distribution of clip duration and class-specific duration variation. Most recordings approach four seconds, while several event-like classes contain substantially shorter examples.

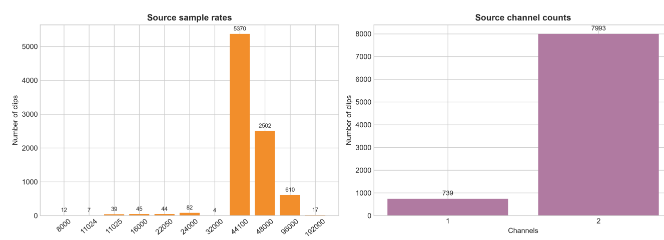

Fig. 4. Source recording characteristics before preprocessing. Sampling rates vary substantially, and most source recordings are stereo.

files were stereo (7,993), while 739 were mono. These format differences are handled explicitly through mono conversion and resampling rather than being allowed to introduce inconsistent model inputs.

The time-domain examples further show that classes differ in temporal structure: engine idling is comparatively periodic, dog barks appear as repeated transient bursts, and gun shots are short high-energy events followed by silence. Such variation motivates a time-frequency representation that can preserve both spectral content and temporal location.

IV. METHODOLOGY

A. Audio Preprocessing

All architectures received the same input representation. Each recording was converted to mono, resampled to 22,050 Hz, and normalized to four seconds. Short recordings were padded with trailing zeros. For recordings longer than the target length, training used a seeded random crop, while validation and test processing used deterministic center cropping. A Log-Mel spectrogram was then computed using a 1,024-

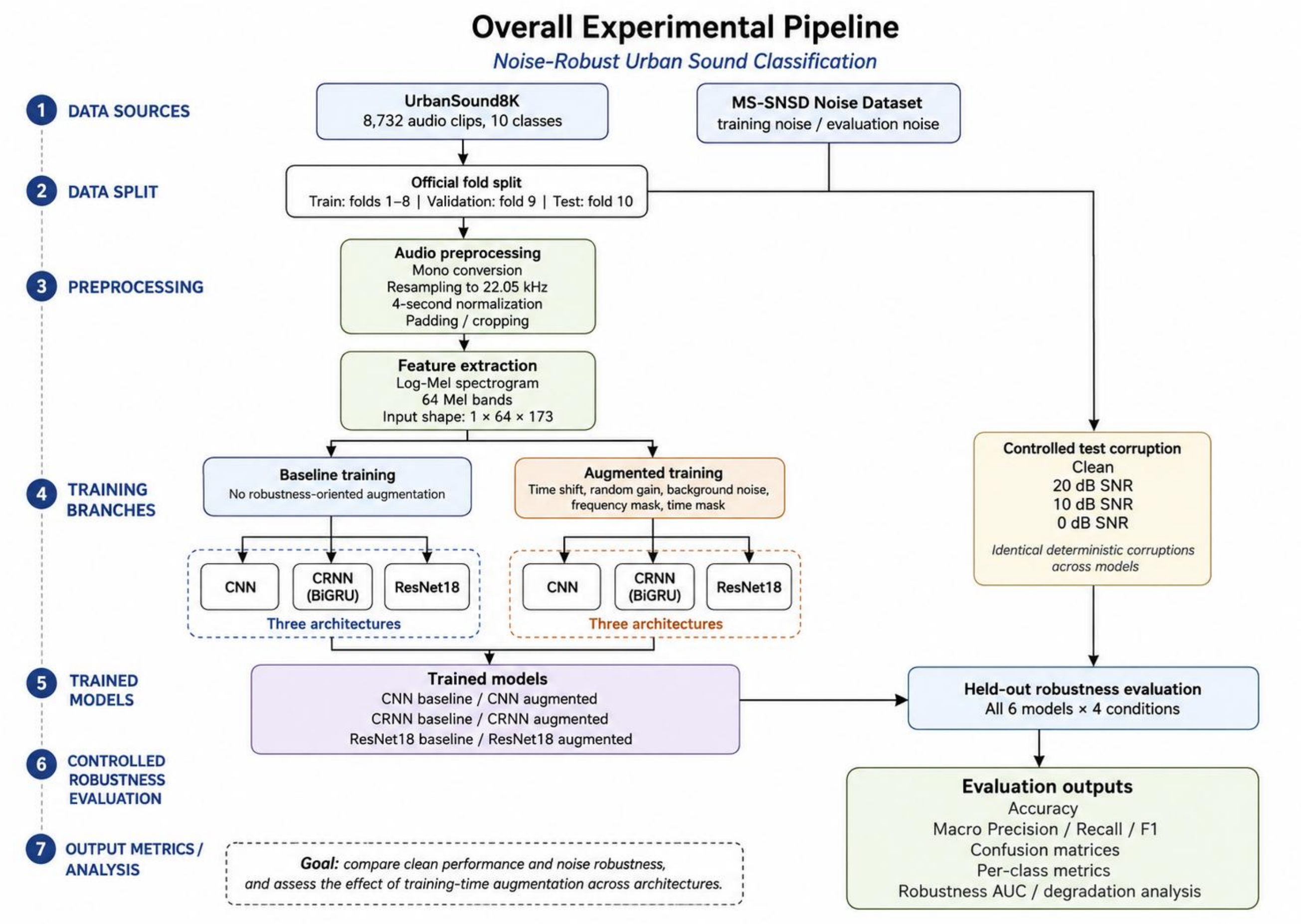

Fig. 5. Overall experimental pipeline. UrbanSound8K is partitioned using the official folds, standardized into Log-Mel spectrograms, and used to train matched baseline and augmented variants of CNN, CRNN, and ResNet18 models. The six trained models are then evaluated on the held-out test fold under identical deterministic clean, 20 dB, 10 dB, and 0 dB conditions generated from the disjoint MS-SNSD evaluation-noise bank.

TABLE III
SHARED PREPROCESSING CONFIGURATION

Setting	Value
Sample rate	22,050 Hz
Clip duration	4.0 s
Input channels	Mono
FFT/window length	1,024
Hop length	512
Mel bands	64
Dynamic range	80 dB
Input shape	$1 \times 64 \times 173$
Normalization	Per-example standardization

sample FFT/window, a 512-sample hop, 64 Mel bands, power 2.0, HTK Mel scaling, and an 80 dB dynamic-range limit. Each spectrogram was standardized independently, producing a model input of size $1 \times 64 \times 173$.

B. Training-Time Augmentation

Two training regimes were defined for each architecture. The baseline condition used the shared preprocessing pipeline without robustness-oriented waveform or spectrogram aug-

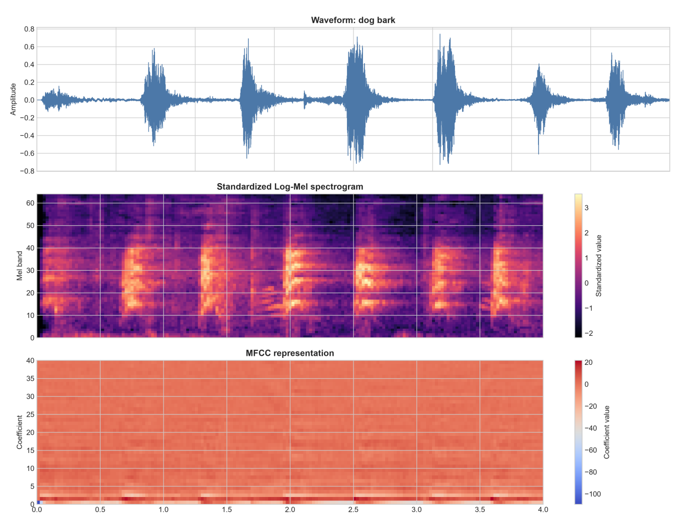

Fig. 6. Representative waveform and derived time-frequency representations for a dog bark after preprocessing. Log-Mel spectrograms are used as the common neural network input.

TABLE IV
MODEL CONFIGURATIONS

Architecture	Batch size	Parameters
CNN	16	94,186
CRNN–BiGRU	16	293,610
ResNet18	8	11,175,370

mentation. In the augmented condition, only training samples were eligible for the following independent transformations: time shift with probability 0.5 and maximum displacement of 20% of the clip length; random gain with probability 0.5 in the range -6 to $+6$ dB; additive background noise with probability 0.5 at a randomly selected SNR between 0 and 20 dB; frequency masking with probability 0.5 and width up to eight Mel bins; and time masking with probability 0.5 and width up to 16 frames. Pitch shifting and time stretching were disabled in the final experiment. Validation and test samples never received training augmentation.

Frequency and time masking follow the general idea of masking contiguous regions in filter-bank features introduced by SpecAugment [7]. Training noise was drawn only from the MS-SNSD training noise directory, whereas final robustness evaluation used a disjoint MS-SNSD test-noise directory [10].

C. Model Architectures

The compact CNN contained three convolutional stages with 32, 64, and 128 channels, followed by adaptive pooling, dropout 0.3, and a ten-class output layer. This model provides a relatively low-capacity spectrogram baseline.

The CRNN used convolutional stages with the same channel progression to extract local time-frequency features. The feature map was then rearranged as a temporal sequence and processed by one bidirectional GRU layer with hidden size 128 and dropout 0.3 before classification. This design combines local spectral pattern recognition with recurrent temporal context [8].

The ResNet18 model used the residual architecture of He *et al.* [9], modified for a one-channel spectrogram input and a ten-class output. It was trained from random initialization rather than ImageNet weights, and dropout 0.2 was applied in the classification path. Thus, the ResNet18 results reflect capacity and residual architecture rather than transfer from an external visual dataset.

D. Model Selection Rationale and Assumptions

The three primary models were selected to compare progressively different ways of modeling the same Log-Mel input. The compact CNN provides a low-parameter local time-frequency baseline, the CRNN adds explicit temporal sequence modeling, and ResNet18 tests whether a substantially deeper residual model benefits differently from the same augmentation policy. A classical MFCC-based classifier such as an SVM was considered as a possible alternative, but it was not used as one of the three primary models because the central comparison concerns architecture-dependent robustness within a shared deep-learning and spectrogram pipeline. MFCCs were retained for exploratory analysis rather than used as the neural input.

The methodology assumes that the supplied UrbanSound8K class labels and predefined folds are suitable supervised targets and grouped partitions; that standardizing recordings to a common sample rate, channel count, and duration preserves sufficient class information; and that the held-out MS-SNSD noise bank provides a controlled, though not exhaustive, proxy for background interference. These assumptions are addressed by preserving all official samples and folds, retaining short events through padding rather than removal, using the same preprocessing for every architecture, and keeping training and evaluation noise banks disjoint. Class imbalance is handled through macro-averaged metrics rather than an assumption of equal class frequency.

E. Controlled Noise Corruption

Final evaluation was performed under four conditions: clean, 20 dB, 10 dB, and 0 dB SNR. For clean waveform x and noise segment n, mean-square powers P_x and P_n were used to scale the noise. For target SNR s in decibels, the scale factor was

$$\alpha = \sqrt{\frac{P_x}{P_n\, 10^{s/10}}}, \tag{1}$$

and the corrupted signal was $x_{\mathrm{noisy}} = x + \alpha n$. This is consistent with

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \left(\frac{P_x}{P_{\alpha n}} \right). \tag{2}$$

A corruption seed of 2025, independent of the training seed, deterministically selected the noise file and segment for each sample. Consequently, every model received the same corruption for a given test sample and SNR. The held-out noise bank contained 51 files, and the evaluator verified that training and evaluation noise banks did not overlap.

F. Evaluation Metrics

The primary metrics were accuracy and macro F1. Macro precision, macro recall, per-class precision/recall/F1, confusion matrices, and per-sample predictions were also saved. For each architecture and training regime, robustness was summarized by the clean-to-noisy metric drop, relative retention, the ordinary-least-squares slope over the finite 0, 10, and 20 dB points, and a normalized SNR area under the curve (AUC). For macro F1 $F(s)$, the normalized AUC was computed by trapezoidal integration over the three finite SNR points and division by the observed 20 dB range:

$$\mathrm{AUC}_{\mathrm{norm}} = \frac{1}{20} \int_0^{20} F(s)\, ds. \tag{3}$$

The clean condition is excluded from this integral because it has no finite SNR.

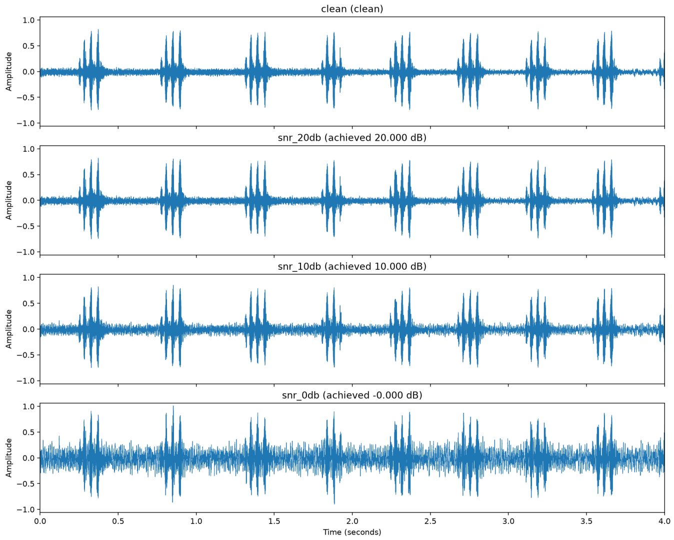

Fig. 7. Example waveform corruption under the clean, 20 dB, 10 dB, and 0 dB evaluation conditions.

V. Experimental Setup

A. Training Protocol

All six configurations used random seed 42 with deterministic execution enabled where practical. Training was limited to 50 epochs and optimized with CrossEntropyLoss and AdamW using a learning rate of 0.001 and weight decay of 0.0001. A ReduceLROnPlateau scheduler reduced the learning rate by a factor of 0.5 after three stagnant epochs. Mixed-precision training was enabled on CUDA. CNN and CRNN used batch size 16, while ResNet18 used batch size 8; each loader used two workers.

Validation macro F1 was the checkpoint-selection metric because the dataset is not perfectly class-balanced. Early stopping used a patience of eight epochs. The best validation checkpoint was stored separately from the latest recovery checkpoint. The six completed training runs comprised CNN baseline, CNN augmented, CRNN baseline, CRNN augmented, ResNet18 baseline, and ResNet18 augmented. Their recorded completed-process runtimes summed to approximately 10.1 h, although interrupted incomplete epochs are not fully represented in that total.

B. Reproducibility and Evaluation Controls

The selected `best.pt` checkpoint from each full run was evaluated once on all 837 fold-10 test samples. No test sample limit was applied. For noisy conditions, the same sample identifiers, target order, selected noise files, noise segments, and achieved SNR values were verified across all six model evaluations. The evaluator rejected recovery-only checkpoints, smoke-run checkpoints, mismatched manifests, and overlapping training/evaluation noise banks. Per-sample outputs recorded the requested and achieved SNR, target, prediction, sample identifier, fold, noise path, and corruption seed. The project's automated verification suite reported 121 passing tests.

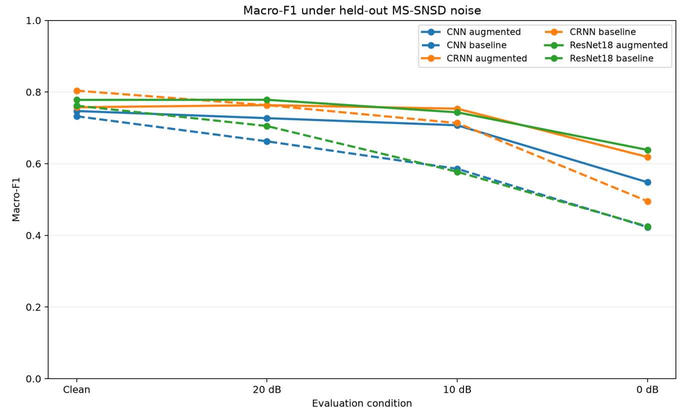

Fig. 8. Macro-F1 performance under clean and controlled background-noise conditions for the six trained configurations.

C. Hardware and Software

Experiments were executed on Windows 10 with Python 3.13.1, PyTorch 2.11.0+cu126, CUDA 12.6, and cuDNN 91002. Training used an NVIDIA GeForce GTX 1660 Ti with approximately 6 GB of GPU memory. The environment was maintained in a repository-local virtual environment, and each experiment saved configuration and environment snapshots together with logs and training histories.

D. Code Availability

The implementation, experiment configurations, training and evaluation scripts, and reproducibility utilities are maintained in the project repository at https://github.com/TASRIF-67/cse437-project. Large generated artifacts, including model checkpoints and some result files, may be excluded from version control because of storage size; the repository documents the commands and configuration structure used to reproduce the reported pipeline.

VI. Results

A. Clean and Noisy-Condition Performance

Table V reports the complete 6×4 test matrix. The architecture-matched baseline-versus-augmented design provides the primary comparative analysis: each model family is evaluated with and without the same robustness-oriented training policy under identical held-out conditions. Under clean audio, the baseline CRNN achieved the highest macro F1, 0.8037, and the highest accuracy, 0.7838. As noise increased, the ranking changed. At 20 dB, augmented ResNet18 achieved the highest macro F1 (0.7783); at 10 dB, augmented CRNN was highest (0.7535); and at 0 dB, augmented ResNet18 was highest (0.6384). The clean leader therefore was not the most robust model under strong interference.

Figure 8 shows the different degradation profiles. All three augmented variants retain more macro F1 than their matched baselines at 0 dB. The effect is especially pronounced for ResNet18, whose baseline macro F1 falls from 0.7617 clean to 0.4249 at 0 dB, whereas the augmented variant retains 0.6384.

TABLE V
FOLD-10 ACCURACY AND MACRO F1 UNDER CONTROLLED NOISE

		Accuracy				Macro F1			
Architecture	Training	Clean	20 dB	10 dB	0 dB	Clean	20 dB	10 dB	0 dB
CNN	Baseline	0.7121	0.6703	0.6057	0.4504	0.7323	0.6624	0.5857	0.4230
CNN	Augmented	0.7288	0.7168	0.6989	0.5496	0.7469	0.7270	0.7071	0.5481
CRNN	Baseline	0.7838	0.7491	0.7085	0.4994	**0.8037**	0.7632	0.7134	0.4952
CRNN	Augmented	0.7419	0.7491	0.7384	0.6141	0.7574	0.7634	**0.7535**	0.6186
ResNet18	Baseline	0.7395	0.6977	0.5938	0.4421	0.7617	0.7050	0.5774	0.4249
ResNet18	Augmented	0.7575	0.7575	0.7264	0.6081	0.7781	**0.7783**	0.7432	**0.6384**

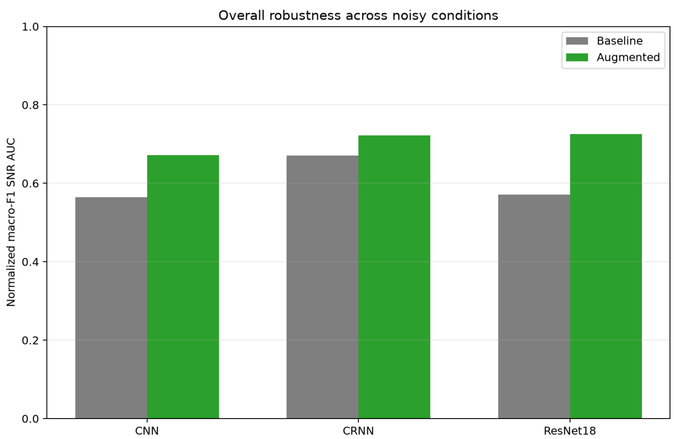

Fig. 9. Normalized macro-F1 SNR area under the noisy-condition curve. Higher values indicate stronger performance across 0, 10, and 20 dB.

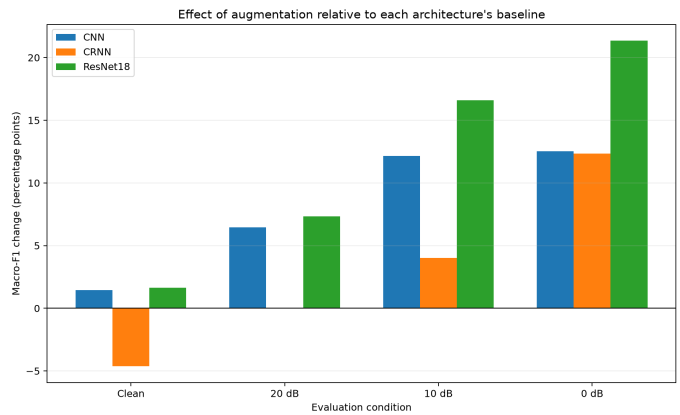

Fig. 10. Architecture-matched macro-F1 difference between augmented and baseline training at each evaluation condition.

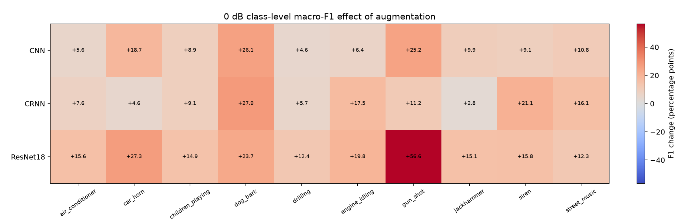

Fig. 11. Per-class F1 change from augmentation at 0 dB for each architecture.

B. Overall Robustness

The normalized SNR-AUC summary in Table VI separates performance across the noisy range from the single clean score. Augmented ResNet18 ranked first with 0.7258, followed by augmented CRNN with 0.7222 and augmented CNN with 0.6724. The three augmented models therefore occupied the top three robustness-AUC positions, while baseline CRNN, despite leading on clean macro F1, ranked fourth by noisy-curve AUC.

C. Effect of Training-Time Augmentation

Architecture-matched differences are shown in Table VII and Fig. 10. Augmentation increased normalized macro-F1 SNR AUC for every architecture: by 0.1082 for CNN, 0.0509 for CRNN, and 0.1546 for ResNet18. At 0 dB, macro F1 increased by 0.1252, 0.1234, and 0.2135, respectively. CNN and ResNet18 also improved slightly on clean data. CRNN showed a different pattern: clean macro F1 decreased by 0.0464, while 0 dB macro F1 increased by 0.1234.

D. Class-Level Behavior at 0 dB

Class-level analysis shows that the aggregate gains were not confined to a single category. For ResNet18, augmentation improved 0 dB F1 for all ten classes. Its largest increases were observed for `gun_shot` (0.2703 to 0.8358), `car_horn` (0.6122 to 0.8852), `dog_bark` (0.4881 to 0.7253), `engine_idling` (0.1308 to 0.3289), and `siren` (0.3883 to 0.5467). Even after augmentation, engine idling, air conditioner, siren, and street music remained comparatively difficult for ResNet18 at 0 dB.

VII. DISCUSSION

A. Effect of Augmentation on Noise Robustness

Across all three architectures, training-time augmentation improved performance retention under additive background noise. Every augmented model achieved both a higher 0 dB macro F1 and a higher normalized macro-F1 SNR AUC than its architecture-matched baseline. The benefit is therefore visible not only at the most severe condition but across the noisy-condition curve. This extends the general augmentation findings reported in prior ESC work [3], [5] by measuring matched models under explicit SNR-controlled test conditions rather than standard held-out performance alone.

TABLE VI
MACRO-F1 ROBUSTNESS SUMMARY

Architecture	Training	Clean F1	0 dB F1	Normalized SNR AUC	AUC rank
CNN	Baseline	0.7323	0.4230	0.5642	6
CNN	Augmented	0.7469	0.5481	0.6724	3
CRNN	Baseline	**0.8037**	0.4952	0.6713	4
CRNN	Augmented	0.7574	0.6186	0.7222	2
ResNet18	Baseline	0.7617	0.4249	0.5712	5
ResNet18	Augmented	0.7781	**0.6384**	**0.7258**	1

TABLE VII
AUGMENTED-MINUS-BASELINE EFFECTS

Architecture	Clean accuracy	Clean macro F1	0 dB accuracy	0 dB macro F1	Macro-F1 AUC
CNN	+0.0167	+0.0146	+0.0992	+0.1252	+0.1082
CRNN	-0.0418	-0.0464	+0.1147	+0.1234	+0.0509
ResNet18	+0.0179	+0.0163	+0.1661	+0.2135	+0.1546

The result is also compatible with the motivation behind time-frequency masking and other perturbation-based training methods: exposure to partial or corrupted evidence during training can encourage a classifier to rely on a wider set of cues [7]. The present experiment does not isolate which individual augmentation caused the gain because the augmented condition combines waveform and spectrogram transformations. The measured improvement should therefore be attributed to the complete augmentation policy rather than to one component.

B. Architecture-Dependent Response

Although the noisy-condition benefit appeared for every architecture, its magnitude and clean-data trade-off differed substantially. ResNet18 obtained the largest increase in normalized SNR AUC (+0.1546) and the largest 0 dB macro-F1 increase (+0.2135). CNN improved more moderately. CRNN became substantially more robust at 0 dB but lost 0.0464 macro F1 on clean audio. Augmentation therefore did not shift all architectures upward by a similar amount.

The architecture-dependent response is consistent with prior observations that augmentation can affect sound classes and model families differently [3], [4]. The current evidence does not establish a mechanistic cause. One plausible interpretation is that the higher-capacity ResNet18 benefits strongly from the additional variation supplied by augmentation, whereas the recurrent model may trade some clean specialization for tolerance to perturbation. This remains a hypothesis because no representation-level or augmentation-ablation analysis was performed.

C. Clean Performance Versus Robustness

A central empirical result is the separation between clean performance and robustness. Baseline CRNN produced the highest clean macro F1 (0.8037), yet augmented ResNet18 produced the best noisy-curve AUC (0.7258) and the best 0 dB macro F1 (0.6384). Selecting a deployment model solely by clean validation or clean test performance could therefore choose a different architecture from one selected for background-noise tolerance. For applications where acoustic conditions are uncertain, robustness-specific evaluation is a useful companion to conventional clean-set metrics.

VIII. LIMITATIONS AND FUTURE WORK

A. Limitations and Threats to Validity

The results are meaningful for the completed controlled experiment but have several important boundaries. First, final evaluation used one official UrbanSound8K fold rather than full ten-fold cross-validation. The chosen split preserved the dataset's fold organization, but a different held-out fold could change absolute scores. Second, each of the six training configurations has one completed training seed. Consequently, run-to-run optimization variance is not quantified, and no result is described as statistically significant.

Third, robustness was evaluated with one held-out MS-SNSD noise collection containing 51 files and three finite noisy SNR levels (0, 10, and 20 dB). These conditions provide a controlled severity sweep but do not represent every possible real acoustic environment. Fourth, the augmented condition combines multiple transformations, so the individual contribution of time shift, gain, noise mixing, time masking, and frequency masking cannot be separated from the current experiment.

Finally, ResNet18 was trained from scratch and modern pretrained audio transformers or large-scale audio encoders were outside the scope of this course project. The conclusions should therefore be interpreted as evidence for these three architectures, this split, and this corruption protocol, rather than as a universal ranking of environmental sound classifiers.

B. Future Work

The most direct extension is to repeat each training condition with multiple random seeds and report uncertainty using paired bootstrap confidence intervals or another paired resampling procedure on the common test predictions. Full ten-fold evaluation would reduce dependence on the single held-out fold, while additional independent noise collections and a denser SNR grid would test whether the observed ranking persists across broader acoustic conditions. An augmentation ablation could separate the contribution of additive noise, time and frequency masking, gain, and temporal shifting. Finally, pretrained audio encoders or transfer-learning variants of residual networks could be compared with the current from-scratch models, provided that external pretraining is reported as a separate experimental factor rather than mixed into the present architecture-matched comparison.

IX. Conclusion

This study compared baseline and augmented training for three environmental sound classifiers and evaluated each selected model under identical clean and SNR-controlled noisy conditions. Across CNN, CRNN, and ResNet18, augmentation increased normalized macro-F1 SNR AUC and improved macro F1 at 0 dB, showing a consistent robustness benefit within the completed held-out-fold experiment. The size of that benefit, however, depended on architecture and was not always accompanied by higher clean performance.

Augmented ResNet18 provided the strongest overall robustness, with normalized macro-F1 SNR AUC 0.7258 and 0 dB macro F1 0.6384, while baseline CRNN remained the strongest clean model at macro F1 0.8037. This difference reinforces the need to evaluate robustness directly rather than infer it from clean accuracy. Within the stated experimental boundaries, the results support augmentation as a practical robustness strategy while also showing that its effect depends on the architecture being trained.

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